Flight

Data Analysis Project



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https://github.com/KumarBoste/Flight_Data_Analysis

Project Report: Flight Data Analysis

1. Project Overview

The **Flight Data Analysis** project is a comprehensive relational database initiative designed to extract operational and business insights from airline industry data. The aviation sector generates large volumes of relational data across flights, ticketing, route networks, airport hubs, and passenger loyalty profiles.

This project structures and interrogates a relational schema encompassing **Airports**, **Airlines**, **Flights**, **Tickets**, and **Passengers** using Structured Query Language (SQL). By leveraging relational joins, aggregate functions, Common Table Expressions (CTEs), and advanced window analytical functions (ROW_NUMBER(), RANK()), the project delivers key operational intelligence—such as hub congestion, carrier dominance, route-level pricing, and customer lifetime value across loyalty tiers.

2. Problem Statement

Aviation management and airline revenue teams frequently face critical operational and commercial challenges:

1. **Airport Congestion & Hub Efficiency:** Inability to pinpoint the busiest departure hubs to optimize slot allocation and gate staffing.
2. **Airline Market Share:** Lack of clarity on ticket volume and carrier presence across major domestic and international hubs.
3. **Flight Route Classification:** Need for operational duration profiling to distinguish short-haul turnarounds from medium- and long-haul operations.
4. **Revenue & Pricing Optimization:** Identifying premium fare ceilings across specific origin-destination pairs.
5. **Customer Loyalty & Value Segmentation:** Determining the highest-value patrons within distinct frequent flyer tiers to drive targeted loyalty initiatives and personalized retention programs.

This project addresses these challenges by transforming raw flight and booking transactions into structured, decision-ready analytics.

3. Analysis Queries

Q1. Business Airport by Number of Departures

Objective: Identify high-density departure hubs to help airport authorities and ground operations optimize slot planning and gate allocation.

SQL

```sql
-- Q1. Find the bussines airport by the number of flights take off:
SELECT A.Name AS Airport,
       COUNT(*) AS TotalFlights
FROM Flights F
INNER JOIN Airports A ON A.AirportID = F.Origin
GROUP BY A.Name
ORDER BY TotalFlights DESC;
```

OUTPUT :

Airport	TotalFlights
Indira Gandhi International Airport	4
Chhatrapati Shivaji Maharaj International Airport	3
Dubai International Airport	3
Heathrow Airport	3
John F. Kennedy International Airport	3

Q2. Total Tickets Sold per Airline

Objective: Evaluate commercial demand and overall booking volume across carriers.

SQL

```sql
-- Q2. Find the total number of tickets sold per airline.
SELECT AL.Name,
       COUNT(*) AS TicketsSold
FROM Tickets T
INNER JOIN Flights F ON  T.FlightID = F.FlightID
INNER JOIN Airlines AL ON F.AirlineID = AL.AirlineID
GROUP BY AL.Name;
```

OUTPUT :

	Name	TicketsSold
▶	Air India	11
	Delta Airlines	16
	Emirates	11
	British Airways	11
	IndiGo	5

Q3. Indigo-Operated Flight Network Details

Objective: Filter operational schedules and origin-destination routes for a specific major carrier (IndiGo).

SQL

```
- Q3. List all flights operated by 'Indigo' with airport names
(origin and destination).
SELECT F.FlightID, A.Name AS OriginAirport, A2.Name AS
DestinationAirport, F.DepartureTime, F.ArrivalTime
FROM  Flights F
INNER JOIN Airlines AL ON F.AirlineID = AL.AirlineID
INNER JOIN Airports A ON F.Origin = A.AirportID
INNER JOIN Airports A2 ON F.Destination = A2.AirportID
WHERE AL.Name ='IndiGo';
```

OUTPUT :

	FlightID	OriginAirport	DestinationAirport	DepartureTime	ArrivalTime
▶	106	Indira Gandhi International Airport	Chhatrapati Shivaji Maharaj International Airport	2025-08-06 07:00:00	2025-08-06 09:00:00
	111	Chhatrapati Shivaji Maharaj International Airport	Indira Gandhi International Airport	2025-08-11 05:30:00	2025-08-11 07:30:00

Q4. Top Operating Airline for Each Airport

Objective: Determine the market leader / dominant carrier at each origin airport.

SQL

```
-- Q4. For each airport, show the top airlines by number of flights
departing from there.
WITH FlightCounts AS(
SELECT F.Origin, F.AirlineID, COUNT(*) AS FlightCount
FROM Flights F
```

```
GROUP BY F.Origin, F.AirlineID
),
RANKEDAirlines AS (
       SELECT *,
              ROW_NUMBER() OVER (PARTITION BY Origin ORDER BY
FlightCount DESC) AS rn
       FROM FlightCounts
)
SELECT A.Name AS AirportName,
          AL.Name AS AirlineName,
       R.FlightCount
FROM RANKEDAirlines R
INNER JOIN Airports A ON R.Origin = A.AirportID
INNER JOIN AIrlines AL ON R.AirlineId = AL.AirlineID
WHERE rn = 1;
```

OUTPUT :

	AirportName	AirlineName	FlightCount
▶	Indira Gandhi International Airport	Air India	3
	Chhatrapati Shivaji Maharaj International Airport	Delta Airlines	1
	Heathrow Airport	British Airways	3
	John F. Kennedy International Airport	Delta Airlines	3
	Dubai International Airport	Emirates	3

Q5. Flight Duration Calculation and Classification

Objective: Classify flight legs into standard operational haul categories: **Short** (less than 2h), **Medium** (2–5h), and **Long** (greater than 5h).

SQL

```
-- Q5. For each flight, show time in hours and categorize it as
Short (<2h), Medium (2-5h), or Long (>5h)
SELECT
    FlightID,
    DepartureTime,
    ArrivalTime,
    TIMESTAMPDIFF(MINUTE, DepartureTime, ArrivalTime) / 60.0 AS
DurationHours,
    CASE
        WHEN TIMESTAMPDIFF(MINUTE, DepartureTime, ArrivalTime) < 120
THEN 'Short'
        WHEN TIMESTAMPDIFF(MINUTE, DepartureTime, ArrivalTime) <=
300 THEN 'Medium'
        ELSE 'Long'
    END AS FlightCategory
FROM Flights;
```

OUTPUT :

FlightID	DepartureTime	ArrivalTime	DurationHours	FlightCategory
101	2025-08-01 06:00:00	2025-08-01 12:00:00	6.0000	Long
102	2025-08-02 10:00:00	2025-08-02 20:00:00	10.0000	Long
103	2025-08-03 05:30:00	2025-08-03 09:30:00	4.0000	Medium
104	2025-08-04 13:00:00	2025-08-04 21:00:00	8.0000	Long
105	2025-08-05 08:00:00	2025-08-05 16:00:00	8.0000	Long
106	2025-08-06 07:00:00	2025-08-06 09:00:00	2.0000	Medium
107	2025-08-07 18:00:00	2025-08-07 22:30:00	4.5000	Medium
108	2025-08-08 03:00:00	2025-08-08 09:00:00	6.0000	Long
109	2025-08-09 14:00:00	2025-08-09 22:00:00	8.0000	Long
110	2025-08-10 11:00:00	2025-08-10 19:00:00	8.0000	Long
111	2025-08-11 05:30:00	2025-08-11 07:30:00	2.0000	Medium
112	2025-08-12 06:45:00	2025-08-12 14:45:00	8.0000	Long
113	2025-08-13 08:00:00	2025-08-13 16:30:00	8.5000	Long
114	2025-08-14 10:15:00	2025-08-14 18:15:00	8.0000	Long
115	2025-08-15 23:30:00	2025-08-16 07:30:00	8.0000	Long
116	2025-08-16 10:00:00	2025-08-16 11:55:00	1.9167	Short

Q6. Passenger Flight Activity & Journey Milestones

Objective: Audit passenger engagement by retrieving first flight date, most recent journey, and lifetime flight count.

SQL

```
-- Q6. Show each passenger's first and last flight dates and
number of flights
WITH RankedTickets AS (
SELECT
       PassengerID,
    MIN(F.DepartureTime) AS FirstFlight,
    MAX(F.DepartureTime) AS LastFlight,
    COUNT(*) AS TotalFlights
FROM Tickets T
INNER JOIN FLights F ON T.FlightID = F.FlightID
GROUP BY PassengerID
)
SELECT
       P.Name,
    R.FirstFlight,
    R.LastFlight,
    R.TotalFlights
FROM RankedTickets R
INNER JOIN Passengers P ON R.PassengerID = P.PassengerID
```

OUTPUT :

Name	FirstFlight	LastFlight	TotalFlights
Aarav Sharma	2025-08-01 06:00:00	2025-08-06 07:00:00	3
Meera Iyer	2025-08-02 10:00:00	2025-08-07 18:00:00	4
Kabir Joshi	2025-08-03 05:30:00	2025-08-08 03:00:00	3
Ananya Reddy	2025-08-04 13:00:00	2025-08-09 14:00:00	3
Rohan Mehta	2025-08-05 08:00:00	2025-08-10 11:00:00	2
Diya Kapoor	2025-08-01 06:00:00	2025-08-06 07:00:00	2
Vikram Sinha	2025-08-02 10:00:00	2025-08-07 18:00:00	2
Sanya Arora	2025-08-03 05:30:00	2025-08-08 03:00:00	2
Arjun Patel	2025-08-04 13:00:00	2025-08-09 14:00:00	2
Neha Das	2025-08-05 08:00:00	2025-08-10 11:00:00	2
Rahul Verma	2025-08-01 06:00:00	2025-08-06 07:00:00	2
Pooja Nair	2025-08-02 10:00:00	2025-08-07 18:00:00	2
Kunal Rao	2025-08-03 05:30:00	2025-08-08 03:00:00	2
Ishita Ghosh	2025-08-04 13:00:00	2025-08-09 14:00:00	2
Manav Bhatia	2025-08-05 08:00:00	2025-08-10 11:00:00	2

Q7. Peak Ticket Price per Origin–Destination Route

Objective: Uncover maximum realized fare prices per route to evaluate premium yield and pricing ceiling.

SQL

```
-- Q7. Find Flights with the highest price tickets sold for each
route (Origin -> Destination)
WITH RouteTickets AS (
SELECT F.FlightID,
            F.Origin,
               F.Destination,
          T.TicketID,
          T.Price,
          RANK() OVER (PARTITION BY F.Origin, F.Destination ORDER BY
T.Price DESC) AS rnk
FROM Tickets T
INNER JOIN Flights F ON T.FlightID = F.FlightID
)
SELECT
        A1.Name AS Origin,
     A2.Name AS Destination,
     RT.Price,
     Rt.TicketID
FROM RouteTickets RT
INNER JOIN Airports A1 ON RT.Origin = A1.AirportID
INNER JOIN Airports A2 ON RT.Destination = A2.AirportID
WHERE rnk = 1
```

OUTPUT :

Origin	Destination	Price	TicketID
Indira Gandhi International Airport	Chhatrapati Shivaji Maharaj International Airport	9700.00	346
Indira Gandhi International Airport	Heathrow Airport	9200.00	321
Chhatrapati Shivaji Maharaj International Airport	John F. Kennedy International Airport	9900.00	352
Chhatrapati Shivaji Maharaj International Airport	Dubai International Airport	9900.00	317
Heathrow Airport	Indira Gandhi International Airport	9200.00	304
Heathrow Airport	Dubai International Airport	9600.00	310
John F. Kennedy International Airport	Heathrow Airport	8300.00	349
John F. Kennedy International Airport	Dubai International Airport	7900.00	345
Dubai International Airport	Indira Gandhi International Airport	8900.00	308
Dubai International Airport	Chhatrapati Shivaji Maharaj International Airport	10400.00	323
Dubai International Airport	Chhatrapati Shivaji Maharaj International Airport	10400.00	353

Q8. Highest-Spending Passenger by Loyalty Status Tier

Objective: Identify the highest monetary value customer within each Frequent Flyer Status category (e.g., Platinum, Gold, Silver).

SQL

```
-- Q8. Find the highest-spending passenger in each Frequent Flyer
Status group
WITH SpendingByPassenger AS (
SELECT
       P.PassengerID,
    P.Name,
    P.FrequentFlyerStatus,
    SUM(T.Price) AS TotalSpent
FROM Passengers P
INNER JOIN Tickets T ON P.PassengerID = T.PassengerID
GROUP BY P.PassengerID, P.Name, P.FrequentFlyerStatus
),
RankedSpending AS (
       SELECT *,
              RANK() OVER (PARTITION BY FrequentFlyerStatus ORDER BY
TotalSpent DESC) AS rn
       FROM SpendingByPassenger
)
SELECT
       Name,
    FrequentFlyerStatus,
    TotalSpent
FROM RankedSpending
WHERE rn = 1
```

OUTPUT :

Name	FrequentFlyerStatus	TotalSpent
Aarav Sharma	Gold	17600.00
Sanya Arora	Gold	17600.00
Ananya Reddy	None	20700.00
Kabir Joshi	Platinum	24700.00
Meera Iyer	Silver	26900.00

4. Conclusion

The **Flight Data Analysis** project demonstrates how relational database design and advanced SQL techniques translate operational and commercial data into strategic business intelligence:

- **Operational Insights:** Hub departure counts (Q1) and haul-type classification (Q5) enable ground handlers, gate crews, and fuel planners to allocate resources efficiently based on turnaround profiles.
- **Market Dominance & Fleet Strategy:** Carrier volume analysis (Q2, Q3) and airport dominance queries (Q4) highlight which airlines anchor regional hubs, aiding airport authorities in route development and competitive benchmarking.
- **Revenue & Pricing Optimization:** Route-level fare analysis (Q7) uncovers peak fare elasticity, allowing revenue management teams to refine dynamic pricing models.
- **Customer Lifetime Value & Retention:** Tracking passenger trip history (Q6) alongside loyalty tier spenders (Q8) gives marketing and loyalty programs clear target segments for high-touch customer retention and tier promotions.

Overall, this project forms an extensible SQL foundation for integrating flight telemetry, scheduling dashboards, and airline business intelligence pipelines.